Justin Xu

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Healthcare Data Scientist (MS Biostatistics) building clinical AI and machine learning systems, from LLM-based phenotyping agents and multimodal deep learning on medical imaging to predictive models for patient risk stratification. Strong foundation in causal inference, survival analysis, and GLM, combined with production-oriented engineering in Python/R/SQL/SAS for deploying reproducible, data-driven solutions in regulated healthcare settings.

EDUCATION

Columbia University in the City of New York, New York, NY — Sept. 2024 – May 2026

Master of Science in Biostatistics — GPA: 3.8/4.0

Relevant Course Topics: Statistical Inference, Statistical Learning, MCMC, GLM, Regression, Modern Analysis, Probability

University of Chicago, Chicago, IL — Sept. 2019 – Mar. 2023

Bachelor of Arts in Biology — GPA: 3.6/4.0

Awards: Jeff Metcalf Fellowship Grant ($5000), Ecology and Evolution Undergraduate Research Fellowship ($5000) (2x)

SKILLS & LANGUAGES

Machine Learning & AI: Predictive Modeling, Deep Learning (Medical Imaging, Foundation Models), LLMs/RAG, NLP, AI Agents, Feature Engineering, Model Evaluation (ROC-AUC, PR, Cross-Validation)
Statistics: Causal Inference & Mediation, Survival Analysis, GLM, ANCOVA, PCA, Bayesian/MCMC, A/B Testing
Programming & Tools: Python (pandas, scikit-learn), R (tidyverse, ggplot2, survival, MatchIt), SQL, SAS, Git, Streamlit, RShiny
Healthcare & Clinical: EHR/Claims Data, Patient Risk Stratification, Clinical Trial Analysis, HIPAA-Aware Workflows
Data Engineering: ETL Pipelines, SQLite, Interactive Dashboards, Reproducible Workflows
Languages: Mandarin (native), Japanese (fluent), Spanish (advanced), French (Beginner)

PROFESSIONAL EXPERIENCE

Cedars-Sinai, Research Assistant, Los Angeles, LA — Aug. 2026 – Present

- Built CohortIQ, an **AI/LLM-based clinical phenotyping system** with an **NLP** rules layer and plain RAG retrieval wrapped by an AI Agent orchestrator with a **streamlit UI** using a 6110-patient patient evaluation population, reducing token usage by 90%
- Created an end-to-end **deep learning** pipeline on breast MRI to predict pCR to neoadjuvant radiotherapy while using **multimodal AI** and CT-pretrained **foundation models** to analyze 1451 pCR-labeled patients' multi-sequence MRI with clinical notes

Columbia University Valeri Lab, Data Science Research Assistant, New York, NY — Apr. 2025 – Aug. 2026

- Engineered an **automated causal-mediation analysis pipeline** in R processing **5,000+ longitudinal patient observations**, enabling 200+ healthcare researchers to model time-varying mediators and cutting analysis time from weeks to days
- Led development of a novel **multi-timepoint mediator function** for the CMAverse **R package** (345 citations, 200+ active users), shipping a full testing suite and published implementation guide adopted into production research workflows

Columbia University Ogden Lab, Research Assistant, New York, NY — Oct. 2024 – Aug. 2026

- Designed and automated a **biomedical image-processing pipeline** analyzing 650+ biological objects (mitochondrial/egg morphology), replacing manual labeling and enabling scalable, high-throughput analysis
- Computed distance matrices to quantify morphological variation, reducing image artifacts by 40%; generated **publication-quality visualizations and statistical reports** to communicate analysis insights

Bristol Myers Squibb, Data Analyst — Phase 3 Major Depressive Disorder Trial, Chicago, IL — Jun. 2023 – Aug. 2023

- Built an interactive **R Shiny dashboard** for a Phase 3 Major Depressive Disorder trial (280 patients), creating an analytics prototype adopted by clinical operations teams as a validation framework for future studies
- Analyzed adjunctive pimavanserin treatment efficacy and produced publication-quality visualizations (e.g., longitudinal plots, bar charts) to communicate treatment-response patterns to cross-functional clinical stakeholders

IQVIA, Data Analyst — Phase 3 Acne Vulgaris Trial, Chicago, IL — Aug. 2023 – Oct. 2023

- Developed an **automated R-based data-cleaning and statistical-analysis pipeline** for a Phase 3 acne vulgaris trial (1,000 subjects), implementing **SAP-compliant ANCOVA and CMH tests** to reduce FDA documentation preparation time by 50%
- Generated evidence documentation that supported regulatory submissions under HIPAA-aware, regulatory-grade rigor

DATA SCIENCE PROJECTS

30-Day Hospital Readmission Prediction, Python, scikit-learn, pandas Jan. 2025 – Jun. 2025

- Built an **end-to-end machine learning pipeline on a 50,000-patient dataset** (demographics, comorbidities, prior utilization) with **clinically grounded feature engineering** (LACE score, Charlson Comorbidity Index, high-utilizer and frailty proxies) to predict 30-day hospital readmissions
- Engineered a **leakage-safe scikit-learn Pipeline** with per-fold preprocessing and 5-fold cross-validation; benchmarked **Logistic Regression and Random Forest** models using ROC-AUC and precision-recall analysis on imbalanced data
- Translated model risk scores into a **business case of ~$620K net annual savings (62% ROI)** by flagging the top-risk discharge cohort for intervention, using feature importance to surface clinically valid drivers

Navi Analytics Agent, Python, SQL, SQLite, Streamlit, LLM tool-calling Jan. 2026 – Jul. 2026

- Building a **reproducible ETL pipeline** that transforms raw mobile-app, agent, and subscription logs into a curated SQLite warehouse (dimension/fact/mart layers) with documented **data-quality profiling** (key uniqueness, nulls, test-traffic handling)
- Developing a **Streamlit dashboard** for DAU, retention, agent-quality, and subscription-conversion KPIs, plus a **grounded LLM Analytics Agent** that answers natural-language metric questions via read-only tool-calling over the warehouse

PUBLICATIONS & PRESENTATIONS

Xu, X., Natale, R. (2024). Correlated evolution of beak and braincase morphology is present only in select bird clades. *Journal of Morphology, 285, e21703.*

Xu, X. (2021). 3D morphometrics of shorebird skulls show a strong signal of semi-independent evolution of beak and braincase. *Annual Meeting of the American Ornithological Society.*